

Are more productive banks always better?

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Summary of results

- Authors estimate Malmquist indices of bank productivity for Indian banks between 2002-2018
- They demonstrate that physical measures of productivity (INT and VA approaches) differ quite significantly from profitability-based measures (Operations approach)
- Broadly MI suggests productivity growth in 2002-10 period led by PSBs, followed by a secular decline
- Bank productivity growth leads asset quality declines

Contributions

Relevant for bank management as well as regulators

- **Main contribution:** Productivity measurement of banks needs to account for quality as well as quantity of output
- It is important to interpret the results obtained from the conventional input-output approach based measures of productivity with great care
- Problem of *bad outputs*, i.e NPLs

Roadmap

- Insightful paper and important work
- Comments focused on methodology
- Are more productive banks truly more efficient?
- Do existing measures of bank productivity measure service provision efficiency?
 1. Problems of productivity measurement in banking
 2. How to deal with NPLs
 3. Some interesting questions

Problems of productivity measurement in banking-I

Method	Input	Output	Time	All	PSB	Pvt
INT	Deposits, Costs	Loans	03-18	1.04%	1.94%	1.39%
VA	Costs	Loans and deposits	03-18	0.56%	0.41%	0.66%
OPs	Costs	Total income	03-18	-1.92%	-0.48%	-2.11%

Key takeaway: OPs approach estimates consistent with a common understanding of the Indian banking sector.

Results of the paper depend on INT and VA approaches accurately measuring bank efficiency, but these measures have some *serious shortcomings*, even if they account for the NPL problem

Problems of productivity measurement in banking-II

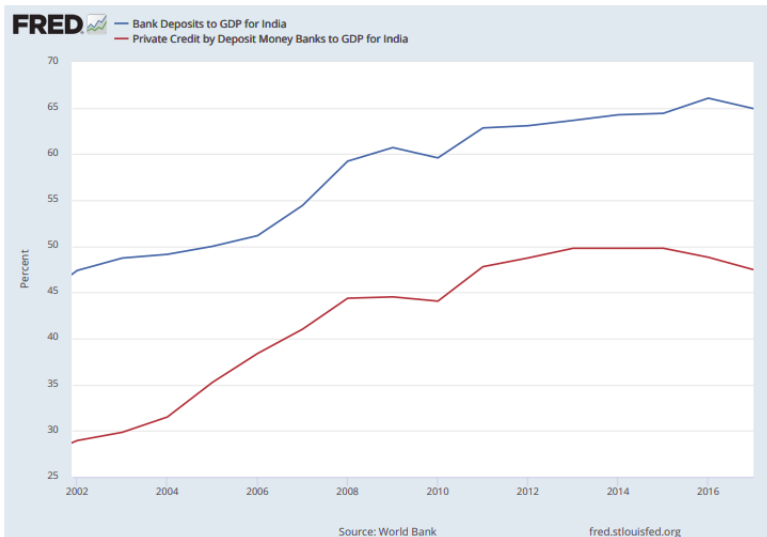
Emphasis on stock variables instead of service flows

	Input	Input type	Output	Output type	Prediction
INT	Costs, Dep	Flow and stock	Loans	Stock	Prod. ↑ when assets grow faster than liabilities and costs
VA	Costs	Flow	Loans and Dep	Stock	Prod. ↑ when output stocks grow faster than costs

Issues: Mixing of stocks and flows, bias due to balancesheet identities.

Ideally in a service industry setting, both inputs and outputs should be service flows.

Macro-aggregates show this prediction clearly



Potential solutions to the measurement problem-I

- Semi-structural approach: Egan, Lewellen and Sunderam (2017,18)
- User cost (imputation) approach: David B. Humphrey (2020)
- Very different papers but similar economic intuition
- Banks have two (service streams) divisions, on the liabilities and asset side with distinct but correlated efficiencies
 1. a deposit-producing division that raises funds by offering consumers services and interest payments
 2. a revenue-producing division that takes funding as an input and converts it into risk-adjusted revenue by making loans and holding securities

Potential solutions to the measurement problem-II

- In Egan et al. deposit productivity is calculated by estimating a consumer demand system for deposits while asset productivity is calculated using an IV regression-based residual approach
- Simple and flexible approach to calculating theory and business-model consistent productivity estimates
- Savings deposit productivity explains over three times as much variation in market-to-book ratios as transaction deposit productivity, and five times as much variation as any other sub-component of productivity

Potential solutions to the measurement problem-III

Move to user cost flow measures if possible

- Humphrey (2020) tries to reconcile productivity measurement of banks with measurement of banking sector income/ output with system of national accounts in the US
- Define bank outputs as value of user cost
- Two sets of costs: Liability side user costs and asset side user costs
- For depositors value of user cost for deposit i is
$$UV_{liabilities,i} = R_{risk.free} - R_{savings.rate,i} * BS \text{ value of liability}_i$$
- For borrowers value of user cost for loan i is
$$UV_{assets,i} = R_{lending,i} - R_{risk.free} * BS \text{ value of loan}_i$$

Potential solutions to the measurement problem-IV

- User cost in theory should provides a more accurate estimate of these bank production outputs (costs) from the user's perspective
- Caveat: It's an approximation to value added plus intermediate inputs to represent asset and liability side output values
- Results from UC/UV approach similar to the operations approach
- Not sure if both these methods are directly applicable in the Indian context, but they are easy to implement

The problem of NPLs

Low hanging fruit..

- Studies in agricultural productivity look at a good output/ bad output framework (Fernandez et al. 2002)
- This has been used in bank productivity literature as well to deal with NPLs as well (Barros et al. 2012)
- These papers generally use a Weighted Russell directional distance model (WRDDM) with banks producing a bad output with Fuiji et al. (2014) being an implementation of it in the Indian context

Some interesting questions

- What are the sources of productivity? How does it vary with bank size and age?
- Many studies have found foreign banks to be most efficient in India, is that the case here as well? Is there any link between FII holdings and bank efficiency?
- How do you account for recapitalisations of PSBs, does productivity improve or fall after recapitalisation?
- How do the results using the OPs approach change when looking at different income sources i.e. Interest vs non-interest income (Ahamed 2016)

Summary

Wishlist

- INT and VA approaches are flawed estimators of bank productivity, OPs approach is far more apt for banking (services)
- Try implementation of comparable “primitive” measures of bank productivity
- Disentangle productivity into its sub-components, service based approach takes service lines as the foundation
- Incorporate NPLs into the estimation of bank productivity using the good-bad output framework
- Important work, hope to read a future version soon!

Thank you

